Radu Micea & Samuel Popa

Task 1:
We ran p-distance on the extracted homologous TP53 locus from NG_017013.2, NC_000017.11 and NC_060941.1:

	NG_017013.2_TP53_5001_24149	NC_000017.11_TP53_7668421_7687490_rev	NC_060941.1_TP53_7572544_7591594_rev
NG_017013.2_TP53_5001_24149	-	0.7430	0.7464
NC_000017.11_TP53_7668421_7687490_rev	-	-	0.3290
NC_060941.1_TP53_7572544_7591594_rev	-	-	-

The closest pair is NC_000017.11 vs. NC_060941.1 because they represent the same human chromosome from different assemblies. The sequence NG_017013.2 is more distant because it is a curated TP53 locus with standardized regions, so it differs more from the raw ones.

Task 2:
NG_017013.2_TP53_5001_24149 vs NC_000017.11_TP53_7668421_7687490_rev

- **Aligned regions**
  - Both alignments use the *same* region of the longer sequence:
    - Seq1: positions 60–19130 (length 19 070) out of 19 149 bp
    - Seq2: positions 0–19070 (length 19 070) out of 19 070 bp
  - This indicates that the entire shorter sequence corresponds to an internal segment of the longer one.
- **Scores**
  - Global alignment score: 19 029.5
  - Local alignment score: 19 070.0
  - In both cases there are 19 070 aligned positions (matches/substitutions), but the global alignment pays extra penalties for gaps, while the local alignment does not introduce any gaps, so it keeps the maximal possible score.
- **Gaps**
  - Global alignment
    - Gaps in seq1 (row 0): 0
    - Gaps in seq2 (row 1): 79

- ■ The global algorithm tries to align the entire length of both sequences, so it adds gap columns at the ends and internally in seq2 to keep everything lined up.

- ○ Local alignment
  - ■ Gaps in seq1: 0
  - ■ Gaps in seq2: 0
  - ■ The best local alignment is a single, long, ungapped stretch where the two sequences are identical.

Small fragment:

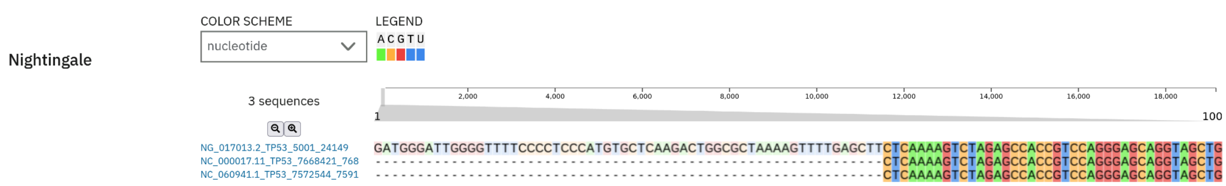
At the end of seq1, the global alignment aligns leading bases of seq1 against gaps in seq2:

```
target GATGGGATTGGGGTTTTCCCCTCCCATGTGCTCAAGACTGGCGCTAAAAGTTTTGAGCTT
       ------------------------------------------------------------
query  ------------------------------------------------------------
```

This region is not part of the local alignment at all; the local alignment starts where the two sequences actually match and then continues as one long ungapped region. A fragment from the middle of the local alignment looks like:

```
target TCTCAAAAAACAAACAAAAAAACAAAACAAAAGAAAACATGGCAAAGCCTTTGAAAGCTT
       ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
query  TCTCAAAAAACAAACAAAAAAACAAAACAAAAGAAAACATGGCAAAGCCTTTGAAAGCTT
```

Task 3:



The highlighted region is conserved across all three TP53 genomic sequences. It probably corresponds to a functionally important element. Because TP53 is a tumor-suppressor gene with essential roles in DNA repair, mutations in conserved positions are strongly selected against, which preserves the sequence.

MSA helps interpretation because it shows which positions are conserved across all sequences simultaneously, not just in pairs. In pairwise alignments, two sequences may look similar in different places, but only an MSA reveals the columns that are consistently identical across the whole group.